Lucas & Prescott (1971)
Investment Under Uncertainty

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Sargent Reading Group

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Motivation

- Explanatory variables in empirical studies of investment:
  - Variables measuring past decisions.
  - Variables measuring current market opportunities.
  - **Variables measuring anticipated, future demand.**
- In Lucas-Prescott (1971):

  Uncertain Future + Adjustment Cost $\Rightarrow$ Investment Under Uncertainty
- Competitive industry equilibrium:
  - product demand shifting stochastically.
  - factor costs remain stable.

A Note on Rational Expectations

- A firm in this industry needs to forecast future demand/prices.

$$Y^e_{t+1} = H(Y_t)$$

- The firm's optimal policy h depends on the belief:

$$y_{t+1} = h(y_t, Y_t, H)$$

- In equilibrium, $Y_t = y_t$. The actual law of motion:

$$Y_{t+1} = h(Y_t, Y_t, H)$$

- Consistency in rational expectations:

$$Y^e_{t+1} = Y_{t+1}$$

▸ Problem Details ▸ REE Definition

Statement of the Model

- Consider an industry consisting of a continuum of small firms.
- Each firm produces a single output q_t using a single input k_t.
- The production function is:

$$0 \leq q_t \leq k_t \tag{1}$$

- Let x_t be gross investment. There are adjustment costs:

$$k_{t+1} = k_t h\left(\frac{x_t}{k_t}\right) \tag{2}$$

  where h is bounded, cont. differentiable, increasing, and strictly concave.

- Assume that $\delta = h^{-1}(1)$ exists and satisfies $0 < \delta < 1$.
  - $x_t = \delta k_t$ is the investment rate that maintains the capital stock.
  - Assume $h(0) > 0$, so even if no investment occurs, some capital will remain.

Statement of the Model II

- Let p_t be the product price. The present value of the firm is:

$$V = \sum_{t=0}^{\infty} \beta^t [p_t q_t - x_t] \tag{3}$$

where $\beta = \frac{1}{1+r}$ and $r > 0$ is the cost of capital.

- Industry demand is subject to random shifts $\{u_t\}$:

$$p_t = D(Q_t, u_t) \tag{4}$$

where D is continuous and strictly decreasing in Q_t and increasing in u_t.

  - $\{u_t\}$ is a Markov process with transition distribution $p(\cdot, u)$
  - $u_{t+1} \in A$ conditional on $u_t = u$ is $\int_A p(dz, u)$.

What does an individual firm actually need to know?

- Denote w_t be the current value of a unit of capital.
- Let w_t^* be the (undiscounted) value per unit expected to prevail next period.
- The current market value of the firm is:

$$w_t k_t = p_t k_t - x + \beta k_t h\left(\frac{x}{k_t}\right) w_t^* \tag{5}$$

- The firm must solve:

$$\max_{x \geq 0} \left[-x + \beta k_t h\left(\frac{x}{k_t}\right) w_t^* \right]$$

- The first-order condition is:

$$0 \geq -1 + \beta h'\left(\frac{x}{k_t}\right) w_t^*, \text{ with equality if } x > 0 \tag{6}$$

- Let $i_t := x_t/k_t$. Combining equations 5 and 6:

$$w_t k_t - p_t k_t = -x_t + k_t \frac{h(i)}{h'(i)}$$

- Dividing by k_t and defining $F(i) := -i + \frac{h(i)}{h'(i)}$:

$$w_t - p_t = F(i)$$

- We need to show that $F'(i) > 0$:

$$F'(i) = -1 + \frac{h'(i)h'(i) - h(i)h''(i)}{[h'(i)]^2} = -1 + 1 - \frac{h(i)h''(i)}{[h'(i)]^2} > 0$$

- Hence, $F(i)$ is invertible. Let $g = F^{-1}$:

$$\boxed{x_t = k_t g(w_t - p_t), \qquad \text{with } g'(\cdot) > 0} \tag{7}$$

Rational Expectations Equilibrium

- An industry equilibrium for a fixed initial state (k_0, u_0) is a triple of sequences $\{q_t^0, x_t^0, p_t^0\}$ such that:
  1. The inverse demand curve $p_t = D(Q_t, u_t)$ holds for every history;
  2. The plan $\{q_t^0, x_t^0\}$ maximizes expected present value

$$\mathbb{E}\left\{\sum_{t=0}^{\infty} \beta^t [p_t^0 q_t - x_t]\right\}$$

     over all plans $\{q_t, x_t\}$ that satisfy $q_t \in [0, k_t]$ and $k_{t+1} = k_t h(x_t/k_t)$ given the price process $\{p_t^0\}$.
  3. $Q_t = q_t$ for every history.

- The $\{p_t^0\}$ that firms take as given is the **same** that their own decisions generate.

How to solve for this equilibrium?

- Fixed point characterization:
  - Let Φ be the composition of two mappings:
    1. Maps a perceived law of motion to a decision rule.
    2. Maps a decision rule to an actual law of motion.
  - Rational expectations equilibrium is a fixed point of Φ.
- However, there is no guarantee that Φ is a contraction.
- In LP ’71: the fixed point is in a space of sequences of functions of histories.

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- **Solution**: Welfare Theorems:
  - Exploit the connection between equilibrium and Pareto optimality.
  - Optimal quantities from the planning problem are REE quantities.
  - Shadow price in the planning problem is the REE price.

Equilibrium as a Planning Problem

- Define consumer surplus as the area under the demand curve:

$$s(q,u) = \int_0^q D(z,u)dz$$

- Define discounted consumer surplus net of investment costs by:

$$S = \mathbb{E}\left\{\sum_{t=0}^{\infty} \beta^t [s(q_t,u_t) - x_t]\right\}$$

- Associated with the problem of maximizing S is the functional equation:

$$v(k,u) = \sup_{x \geq 0}\left\{s(k,u) - x + \beta \int v(kh(x/k),z)p(dz,u)\right\} \tag{8}$$

Theorem 1: Existence and Uniqueness of Equilibrium

- The functional equation (8) has a unique bounded solution v.
- For each (k, u), the supremum is attained at a unique $x(k, u)$.
- In terms of that policy function, the unique industry equilibrium, given (k_0, u_0), is:

$$x_t = x(k_t, u_t), \quad k_{t+1} = k_t h\left(\frac{x(k_t, u_t)}{k_t}\right), \quad q_t = k_t, \quad p_t = D(q_t, u_t)$$

- Proof has two halves:
  1. It shows that competitive equilibrium maximizes S, and conversely.
  2. It shows that the planning problem is solved through Equation (8). Details
- The equilibrium is recursive: the pair (k_t, u_t) is a Markov process.

Long Run Equilibrium with Independent Errors

- Consider the iid case where u_t and u_s are independent for $s \neq t$.
- The functional form in Equation (8) becomes:

$$v(k,u) = \max_{x \geq 0} \left\{ s(k,u) - x + \beta \int v(kh(x/k),z)p(dz) \right\} \tag{9}$$

- Hence, optimal investment does not depend on the stochastic u:

$$x(k,u) = x(k)$$

- The time path of capital stock will be **deterministic**, following:

$$k_{t+1} = k_t h\left(\frac{x(k_t)}{k_t}\right) \tag{10}$$

Theorem 2

- Under independence of the $\{u_t\}$ process, there are two possibilities for the capital stock:
  1. If $k_0 > 0$ and
  $$\int D(0,u)p(du) > \delta + \frac{r}{h'(\delta)}, \tag{11}$$
  then, k_t converges monotonically to the unique stationary value k^c given implicitly by:
  $$\int D(k^c,u)p(du) = \delta + \frac{r}{h'(\delta)} \tag{12}$$
  2. Otherwise, capital converges monotonically to zero.

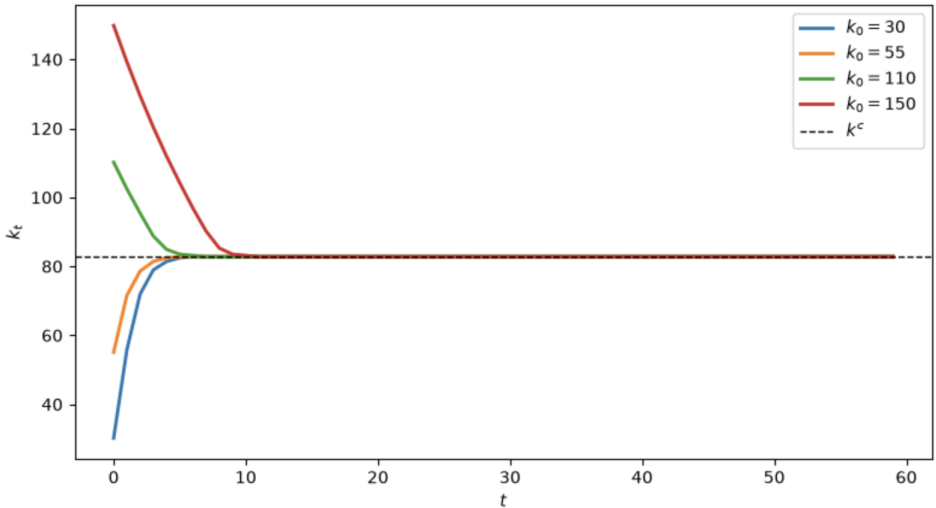


Fig. 114.2 Capital paths under IID demand

Long Run Equilibrium with Serially Dependent Errors

- Demand shifts are positively correlated.
- We need to impose some restrictions on the $\{u_t\}$ process Details
  - Assure that (k_t, u_t) will converge to a stationary distribution independent of (k_0, u_0).
- A Gaussian AR(1) process with $\rho \in (0, 1)$ would satisfy the restrictions:

$$u_{t+1} = \rho u_t + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma^2)$$

Bounding Investment Policies

- Let $\bar{v}(k)$ be the limit of the expected value function as $u \to \infty$
  - $\bar{x}(k)$ be the associated investment policy.

$$\bar{v}(k) = \lim_{u \to \infty} \int v(k, z) p(dz, u)$$

- Let $\underline{v}(k)$ be the limit of the expected value function as $u \to -\infty$
  - $\underline{x}(k)$ be the associated investment policy.

$$\underline{v}(k) = \lim_{u \to -\infty} \int v(k, z) p(dz, u)$$

- Because investment is increasing in u:

$$\underline{x}(k) \leq x(k, u) \leq \bar{x}(k) \tag{13}$$

Bounding the Capital Stock

- Let the lower bound $\underline{k}$ satisfy:

$$\underline{x}(\underline{k}) = \delta \underline{k} \tag{14}$$

- Let the upper bound $\bar{k}$ satisfy:

$$\bar{x}(\bar{k}) = \delta \bar{k} \tag{15}$$

- So capital is bounded $k \in [\underline{k}, \bar{k}]$.

- Quick proof:
  - Consider capital at the lower boundary $k_t = \underline{k}$.
  - From Equation 13:

$$x(\underline{k}, u_t) \geq \underline{x}(\underline{k}) = \delta \underline{k}$$

  - Hence,

$$k_{t+1} = \underline{k} h\left(\frac{x(\underline{k}, u_t)}{\underline{k}}\right) \geq \underline{k} h(\delta) \quad \Rightarrow \quad k_{t+1} \geq \underline{k}$$

Theorem 3

- Lucas-Prescott then prove that:
  1. the sets $(0, \underline{k})$ and $(\bar{k}, \infty)$, paired with any demand state, are **transient**: once the industry leaves them it does not return, and from any starting point it enters $(\underline{k}, \bar{k})$ with probability approaching one.
  2. the set $B = (\underline{k}, \bar{k}) \times E$, where E is the set of possible values of u, is a **single ergodic set**.

- if B is non-empty, then for all (k, u) and every initial state (k_0, u_0),

$$\lim_{t \to \infty} \mathbb{P}\{k_t \leq k, u_t \leq u \mid k_0, u_0\} = P(k, u)$$

  exists and does not depend on (k_0, u_0).
- The function P is a probability distribution that assigns probability zero to the transient sets and positive probability to every subset of B with positive area.

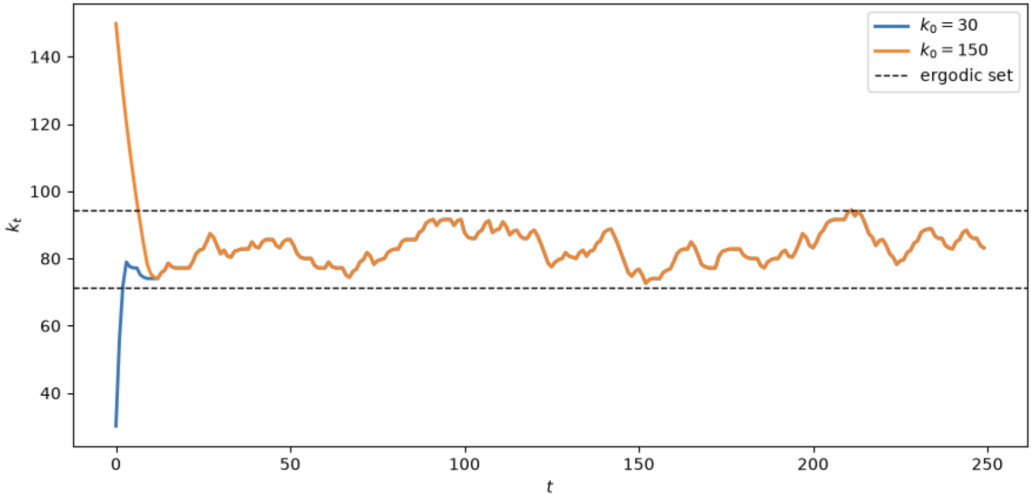


Fig. 114.3 Capital paths entering the ergodic set

Theorem 4 & 5

- If B is non-empty, then for any initial state, with probability one:

$$\lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} k_t = k^*$$

where k^* is the mean of k under the invariant distribution of P.

- The ergodic set is non-empty if and only if:

$$\lim_{u\to\infty} D(0,u) > \delta + \frac{r}{h'(\delta)}$$

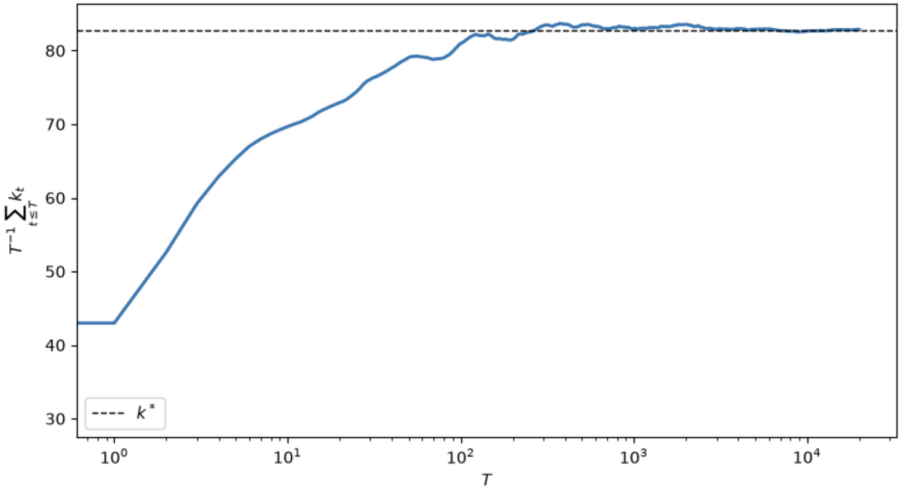


Fig. 114.5 Time average of capital

Conclusion

- This paper complements the investment theory by adding uncertainty.
- Solved for the rational expectations equilibrium using planner's problem.
- With independent shocks, capital stock evolves deterministically.
- With serially correlated shocks, capital stock converges to a single ergodic distribution.

Appendix

Simple Firm's Problem

- The price p_t of the good lies on the inverse demand curve:

$$p_t = a_0 - a_1 Y_t$$

  where
  - $a_i > 0$ for $i = 0, 1$
  - $Y_t = \int_0^1 y_t(\omega) d\omega = y_t$

- Let v be the optimal value function for the firm's problem given H:

$$v(y, Y) = \max_{y'} \left\{ a_0 y - a_1 y Y - \frac{\gamma(y' - y)^2}{2} + \beta v(y', H(Y)) \right\}$$

Back

Definition of a Rational Expectations Equilibrium

Definition 113.1

A **rational expectations equilibrium** or **recursive competitive equilibrium** of the model with adjustment costs is a decision rule h and an aggregate law of motion H such that

1. Given belief H, the map h is the firm's optimal policy function.
2. The law of motion H satisfies $H(Y) = h(Y, Y)$ for all Y.

Back

Operator T

- They use the operator T:

$$Tf(k,u) = \sup_{x \geq 0} \left\{ s(k,u) - x + \beta \int f\big(kh(x/k), z\big) p(dz,u) \right\}$$

- Verify that T is monotone and satisfies a discounting property, so it has a unique fixed point (Blackwell, 1965).
- They show that T preserves concavity and monotonicity in k, which delivers a unique and continuous policy function $x(k,u)$.

Back

Restrictions in $\{u_t\}$ Process

- We need to impose some restrictions on the $\{u_t\}$ process:
  1. From any current u_t, u_{t+1} lands in any non-degenerate interval with positive probability.
  2. u_t has a limiting distribution that does not depend on u_0 and that puts positive probability on every non-degenerate interval.
  3. $\mathbb{P}\{u_{t+1} \geq x \mid u_t\}$ is strictly increasing in u_t.
     - High demand today always signals high demand tomorrow.
  4. Consumer surplus $s(k, u)$ converges uniformly as $u \to \pm\infty$.

Back